

ECO387C - Fall 2024

Problem Set 3 Solutions

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Question 1

- a First, consider the constraint: $0 \leq c_t \leq g_t$. This will not bind at either boundary. If $c_t = 0$ at some t , the CRRA function will yield an infinite marginal utility. If $c_t = g_t$, this means $c_{t+1} = 0$, and this will also lead to an infinite marginal utility. Thus, it is optimal for the agent to always choose $c_t^* \in (0, g_t)$ and this constraint can be disregarded from the optimization. (You can verify this formally by using KKT conditions)

Second, we can consolidate the constraint:

$$g_{t+1} = g_0 - \sum_{\tau=0}^t c_{\tau}$$

This implies

$$g_0 = g_{t+1} + \sum_{\tau=0}^t c_{\tau} \Rightarrow g_0 = \sum_{\tau=0}^{\infty} c_{\tau}$$

Then, we could set up the Lagrangian for the agent:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_t) + \lambda \left[g_0 - \sum_{\tau=0}^{\infty} c_{\tau} \right]$$

- b The first-order conditions of this problem are:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_t} &= 0 \Rightarrow \beta^t u'(c_t) = \lambda \\ \frac{\partial \mathcal{L}}{\partial c_{t+1}} &= 0 \Rightarrow \beta^{t+1} u'(c_{t+1}) = \lambda \end{aligned}$$

The intertemporal Euler equation is

$$u'(c_t) = \beta u'(c_{t+1})$$

c The value function at time 0 is

$$V(g_0) = \max_{\{c_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

We can rewrite this as follows.

$$\begin{aligned} V(g_0) &= \max_{\{c_0\}} u(c_0) + \max_{\{c_t\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^t u(c_t) \\ &= \max_{\{c_0\}} u(c_0) + \beta \max_{\{c_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_{t+1}) \\ &= \max_{\{c_0\}} u(c_0) + \beta V(g_1) \quad \text{s.t.} \quad g_1 = g_0 - c_0 \end{aligned}$$

The time t value function is

$$V(g_t) = \max_{\{c_t\}} u(c_t) + \beta V(g_{t+1}) \quad \text{s.t.} \quad g_{t+1} = g_t - c_t$$

In conventional notation,

$$V(g) = \max_{\{c\}} u(c) + \beta V(g'), \quad \text{s.t.} \quad g' = g - c$$

d Plug the constraint into the objective function:

$$V(g) = \max_{\{c\}} u(g - g') + \beta V(g')$$

The first order condition w.r.t g' is

$$-u'(g - g') + \beta V'(g') = 0$$

We invoke the envelope condition to obtain $V'(g')$. Differentiate the objective function w.r.t g and update one period ahead.

$$V'(g) = u'(g - g') \Rightarrow V'(g') = u'(g' - g'')$$

Combining with the F.O.C above gives us the same Euler equation.

$$u'(g - g') = \beta u'(g' - g'') \Rightarrow u'(c_t) = \beta u'(c_{t+1})$$

e Given CRRA utility, $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, the Euler equation is

$$c_t^{-\gamma} = \beta c_{t+1}^{-\gamma} \Rightarrow \frac{c_{t+1}}{c_t} = \beta^{1/\gamma}$$

The growth rate of consumption is $0 < \beta^{1/\gamma} < 1$. This implies $c_t = \beta^{t/\gamma} c_0$. Plugging this into the budget constraint gives

$$\begin{aligned} g_0 &= \sum_{\tau=0}^{\infty} c_{\tau} \\ &= c_0(1 + \beta^{1/\gamma} + \beta^{2/\gamma} + \dots) \\ &= \frac{c_0}{1 - \beta^{1/\gamma}} \end{aligned}$$

The optimal consumption and saving is therefore

$$c_t^* = (1 - \beta^{1/\gamma})g_t \quad g_{t+1}^* = \beta^{1/\gamma} g_t$$

f The optimal consumption and saving implies

$$c_t = (1 - \beta^{1/\gamma})g_t = (1 - \beta^{1/\gamma})\beta^{t/\gamma} g_0$$

Time-0 value function is

$$V(g_0) = \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma}$$

Plugging c_t into above gives

$$V(g_0) = \frac{g_0}{1-\gamma} (1 - \beta^{1/\gamma})^{-\gamma} = u(g_0)(1 - \beta^{1/\gamma})^{-\gamma}$$

Then it immediately follows that

$$V(g_t) = u(g_t)(1 - \beta^{1/\gamma})^{-\gamma}$$

Question 2

Part a

Note that $\log(X_t/\bar{X}_t) \approx \frac{X_t - \bar{X}_t}{\bar{X}_t} = \widehat{X}_t$

a) Detrending gives

$$\widehat{aX}_t = \frac{aX_t - a\bar{X}_t}{a\bar{X}_t} = \frac{X_t - \bar{X}_t}{\bar{X}_t} = \widehat{X}_t$$

b) Taking logs and detrending gives

$$\widehat{X}_t^a = \frac{X_t^a - \bar{X}_t^a}{\bar{X}_t^a} = \log(X_t^a/\bar{X}_t^a) = a \log(X_t/\bar{X}_t) = a\widehat{X}_t$$

c) Taking logs and detrending gives

$$\widehat{\bar{X}_t \bar{Y}_t} = \frac{X_t Y_t - \bar{X}_t \bar{Y}_t}{\bar{X}_t \bar{Y}_t} = \log(X_t Y_t) - \log(\bar{X}_t \bar{Y}_t) = \log(X_t / \bar{X}_t) + \log(Y_t / \bar{Y}_t) = \widehat{X}_t + \widehat{Y}_t$$

d) Taking logs and detrending gives

$$\frac{\widehat{X}_t}{\widehat{Y}_t} = \frac{\frac{X_t}{\bar{Y}_t} - \frac{\bar{X}_t}{\bar{Y}_t}}{\frac{\bar{X}_t}{\bar{Y}_t}} = \log(X_t / Y_t) - \log(\bar{X}_t / \bar{Y}_t) = \log(X_t / \bar{X}_t) - \log(Y_t / \bar{Y}_t) = \widehat{X}_t - \widehat{Y}_t$$

e) Detrending gives:

$$\widehat{\bar{X}_t + \bar{Y}_t} = \frac{X_t + Y_t - (\bar{X}_t + \bar{Y}_t)}{\bar{X}_t + \bar{Y}_t} = \frac{\bar{X}_t}{\bar{X}_t + \bar{Y}_t} \frac{(X_t - \bar{X}_t)}{\bar{X}_t} + \frac{\bar{Y}_t}{\bar{X}_t + \bar{Y}_t} \frac{(Y_t - \bar{Y}_t)}{\bar{Y}_t} = \frac{\bar{X}_t}{\bar{X}_t + \bar{Y}_t} \widehat{X}_t + \frac{\bar{Y}_t}{\bar{X}_t + \bar{Y}_t} \widehat{Y}_t$$

f) Detrending gives:

$$\widehat{\bar{X}_t - \bar{Y}_t} = \frac{X_t - Y_t - (\bar{X}_t - \bar{Y}_t)}{\bar{X}_t - \bar{Y}_t} = \frac{\bar{X}_t}{\bar{X}_t - \bar{Y}_t} \frac{(X_t - \bar{X}_t)}{\bar{X}_t} + \frac{\bar{Y}_t}{\bar{X}_t - \bar{Y}_t} \frac{(Y_t - \bar{Y}_t)}{\bar{Y}_t} = \frac{\bar{X}_t}{\bar{X}_t - \bar{Y}_t} \widehat{X}_t + \frac{\bar{Y}_t}{\bar{X}_t - \bar{Y}_t} \widehat{Y}_t$$

Part b

a) Applying the formulas above:

$$\widehat{Y}_t = \frac{\bar{C}}{\bar{C} + \bar{I} + \bar{G}} \widehat{C}_t + \frac{\bar{I}}{\bar{C} + \bar{I} + \bar{G}} \widehat{I}_t + \frac{\bar{G}}{\bar{C} + \bar{I} + \bar{G}} \widehat{G}_t$$

b) Applying the formulas above:

$$\widehat{K}_{t+1} = \frac{\bar{I}}{\bar{I} + (1 - \delta)\bar{K}} \widehat{I}_t + \frac{(1 - \delta)\bar{K}}{\bar{I} + (1 - \delta)\bar{K}} \widehat{K}_t$$

Note that at the constant trend, $K_{t+1} = K_t = \bar{K}$, this implies $\bar{I} = \delta\bar{K}$. Then, the linearized equation can be simplified as follows:

$$\widehat{K}_{t+1} = \delta \widehat{I}_t + (1 - \delta) \widehat{K}_t$$

c) Applying the formulas above:

$$\widehat{Q}_t = -\gamma(\widehat{C}_{t+1} - \widehat{C}_t)$$

d) Applying the formulas above:

$$\widehat{W}_t - \widehat{P}_t = \widehat{A}_t - (1 - \alpha)\widehat{N}_t$$